

DASHBALBAR, Mongolia

WMO: 442560

Lat: 49.5508N

Lon: 114.4042E

Elev: 705

StdP: 93.14

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.7	-31.1	-35.8	0.1	-33.0	-33.4	0.2	-30.7	12.1	-17.9	10.4	-15.9	0.4	270	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	31.9	19.9	29.7	19.3	27.6	18.6	21.7	29.3	20.5	27.3	19.5	25.6	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	15.2	25.7	18.2	14.3	23.9	17.3	13.4	22.8	67.4	29.2	62.7	27.5	59.0	25.7	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.4	9.6	-36.6	35.4	3.2	2.7	-38.9	37.3	-40.7	38.9	-42.5	40.3	-44.8	42.3		
(5)			-36.8	23.3	3.1	1.6	-39.1	24.5	-40.9	25.5	-42.7	26.4	-44.9	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-20.8	-16.4	-7.4	2.9	10.8	17.5	20.4	18.0	11.1	1.3	-10.5	-18.3		
	DBStd	15.26	5.32	5.67	6.26	5.06	4.46	3.95	2.92	3.55	4.47	5.10	6.47	5.38		
	HDD10.0	4305	956	739	541	219	44	2	0	1	39	273	614	878		
	HDD18.3	6553	1214	972	799	462	238	59	11	48	221	528	864	1137		
	CDD10.0	948	0	0	0	7	68	228	322	248	71	3	0	0		
	CDD18.3	155	0	0	0	0	4	36	74	38	3	0	0	0		
	CDH23.3	1939	0	0	0	7	107	551	774	429	71	0	0	0		
	CDH26.7	664	0	0	0	1	34	195	275	143	16	0	0	0		
(14) Wind	WSAvg	3.9	3.2	3.4	4.2	5.3	5.4	3.9	3.4	3.5	3.9	4.0	3.6	3.4		
(15) Precipitation	PrecAvg	269	2	2	3	5	19	45	84	65	29	9	3	4		
	PrecMax	502	9	7	9	26	74	181	249	165	75	37	17	8		
	PrecMin	116	0	0	0	0	0	4	18	15	1	1	0	1		
	PrecStd	97	2	2	2	6	15	36	57	35	18	9	4	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.0	2.8	12.7	23.7	30.7	34.9	35.6	33.3	29.0	20.1	8.5	-2.6		
		MCWB	-6.8	-1.3	5.8	13.4	15.8	20.3	22.4	21.7	17.6	10.7	3.7	-4.7		
	2%	DB	-8.1	-1.0	9.2	18.7	26.2	31.2	32.1	30.3	25.3	15.8	4.5	-5.7		
		MCWB	-9.3	-3.4	3.3	9.8	14.7	19.1	20.2	20.1	16.0	8.7	0.8	-7.2		
	5%	DB	-10.6	-4.1	5.6	15.3	22.9	28.8	29.5	27.8	22.5	12.4	1.5	-8.1		
		MCWB	-11.6	-5.8	1.1	7.9	13.3	18.3	20.0	19.1	14.6	6.8	-1.2	-9.3		
	10%	DB	-12.8	-7.0	2.7	12.4	20.3	26.4	27.4	25.4	19.9	9.8	-0.9	-10.5		
		MCWB	-13.7	-8.3	-1.0	6.3	12.0	17.3	19.3	18.1	13.1	5.0	-3.0	-11.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.8	-0.9	6.5	14.2	18.5	22.9	24.0	22.8	18.8	11.4	4.5	-4.3		
		MCDB	-5.5	2.1	12.9	22.4	28.3	32.9	32.7	31.6	27.2	18.7	8.2	-2.7		
	2%	WB	-9.3	-3.5	3.5	10.7	15.7	20.5	22.1	21.1	16.9	8.8	1.4	-7.1		
		MCDB	-8.1	-1.2	8.9	17.9	24.1	28.7	29.1	28.5	23.5	15.2	4.8	-5.8		
	5%	WB	-11.6	-5.8	1.3	8.3	13.9	19.1	20.9	19.8	15.3	7.1	-0.5	-9.2		
		MCDB	-10.7	-4.1	5.5	14.4	21.8	26.6	27.5	26.0	20.8	12.1	2.0	-8.1		
	10%	WB	-13.6	-8.3	-0.9	6.5	12.4	18.0	19.9	18.8	13.7	5.5	-2.4	-11.4		
		MCDB	-12.9	-7.0	2.5	12.1	19.5	25.0	25.7	24.0	18.8	9.5	-0.2	-10.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	13.4	13.5	13.9	14.4	13.4	11.6	11.9	13.3	12.4	11.1	10.5		
		MCDBR	12.6	15.7	16.9	18.8	19.4	17.8	15.8	16.3	18.1	16.7	13.6	11.4		
		MCWBR	11.6	13.6	12.1	12.2	10.9	8.6	7.2	8.0	10.1	10.8	9.9	10.3		
	5% WB	MCDBR	12.5	15.7	16.5	17.6	17.6	15.8	13.6	13.9	15.7	15.8	13.2	11.4		
		MCWBR	11.5	13.7	12.1	12.1	10.8	8.6	7.0	7.4	9.0	10.6	10.2	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.254	0.280	0.320	0.396	0.476	0.457	0.423	0.386	0.337	0.305	0.276	0.261		
	taud		2.274	2.240	2.221	2.084	1.898	2.077	2.250	2.329	2.405	2.409	2.327	2.225		
	Ebn at Noon		804	865	884	843	786	807	832	848	861	830	767	728		
	Edh at Noon		69	93	114	147	186	156	130	114	95	79	66	64		
(44) All-Sky Solar Radiation	RadAvg		1.74	2.96	4.44	5.48	6.21	6.47	5.98	5.37	4.32	2.93	1.78	1.31		
	RadStd		0.09	0.13	0.26	0.33	0.32	0.46	0.37	0.35	0.24	0.13	0.09	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	-1.91	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page